

Unit 7

Comparing More Than Two Groups

Lecture notes for IB DP Biology and Chemistry

Course purpose	To help students move from raw experimental data to cautious, justified scientific claims.
Unit focus	One-way ANOVA, the F-statistic, multiple comparisons, post-hoc tests, effect size, assumptions, and cautious reporting.
Central habit	Do not run many t-tests until one becomes significant. Compare the whole design first, then ask where the differences are.

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1 Purpose of Unit 7

Many IB investigations compare more than two conditions:

- plant growth under four fertilizer concentrations;
- enzyme activity at several temperatures;
- reaction rate under several substrate concentrations;
- diffusion distance in gels with different agar concentrations;
- bacterial survival under several antimicrobial treatments.

The weak approach is to compare every pair of groups with separate t-tests. The stronger approach is to begin with a model for the full experiment.

Core thesis

When there are three or more groups, the first question is usually whether the group factor explains enough variation compared with natural within-group variation. ANOVA is the standard tool for this question.

By the end of this unit, students should be able to:

1. explain why many separate t-tests increase the false-positive risk;
2. state the null and alternative hypotheses for one-way ANOVA;
3. explain the F-statistic as a ratio of between-group variation to within-group variation;
4. read a simple ANOVA table;
5. distinguish statistical significance from biological or chemical importance;
6. report effect size using η^2 or ω^2 ;
7. explain why post-hoc tests are needed after a significant ANOVA;
8. choose cautious wording for IB-style conclusions;
9. recognize when Welch ANOVA, Kruskal-Wallis, or permutation methods may be better than classical ANOVA.

2 Why not just run many t-tests?

Suppose an experiment has four groups:

Control, Low dose, Medium dose, High dose.

There are six possible pairwise comparisons:

Control–Low Control–Medium Control–High Low–Medium Low–High Medium–High

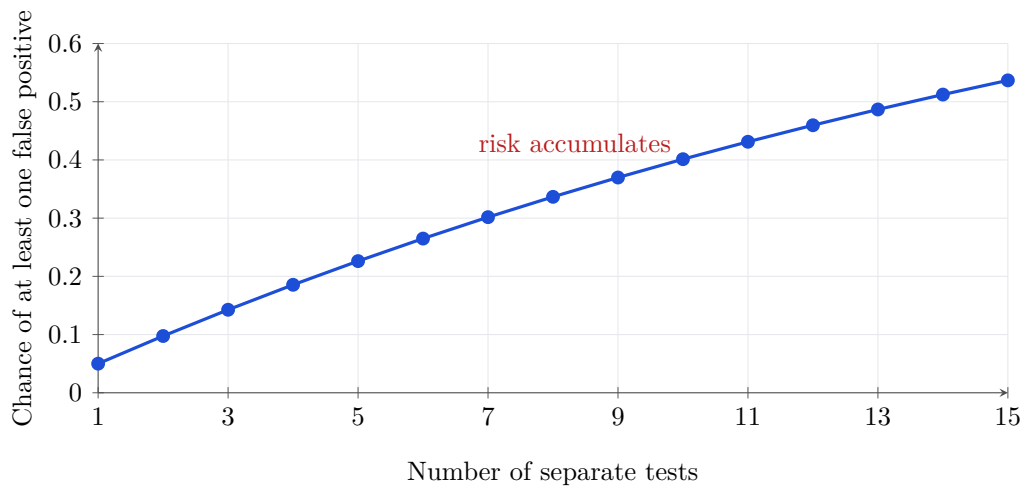
If each test uses $\alpha = 0.05$, each individual test has a 5% Type I error rate under its own null hypothesis. But the chance of getting at least one false positive across several tests is larger than 5%.

Family-wise error

If m independent tests are performed, and each uses $\alpha = 0.05$, the approximate chance of at least one false positive is:

$$P(\text{at least one false positive}) = 1 - (1 - 0.05)^m.$$

The independence assumption is simplified, but the warning is still useful: more tests create more opportunities for false discoveries.

**Bad workflow**

Do not test Control vs Low, Control vs Medium, Control vs High, and so on until something gives $p < 0.05$. That is not careful inference; it is a recipe for accidental significance.

3 The one-way ANOVA question

ANOVA means **analysis of variance**. The name is slightly misleading for students: ANOVA is often used to compare means, but it does so by comparing sources of variation.

For a one-way ANOVA, there is one categorical explanatory variable, usually called a **factor**. Each value of the factor is a **level**.

Example

Research question: Does fertilizer concentration affect plant height after 14 days?

- Response variable: plant height in cm.
- Factor: fertilizer concentration.
- Levels: Control, Low, Medium, High.

The null hypothesis is:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \cdots = \mu_k$$

where k is the number of groups.

The alternative hypothesis is not that all groups differ from each other. It is weaker:

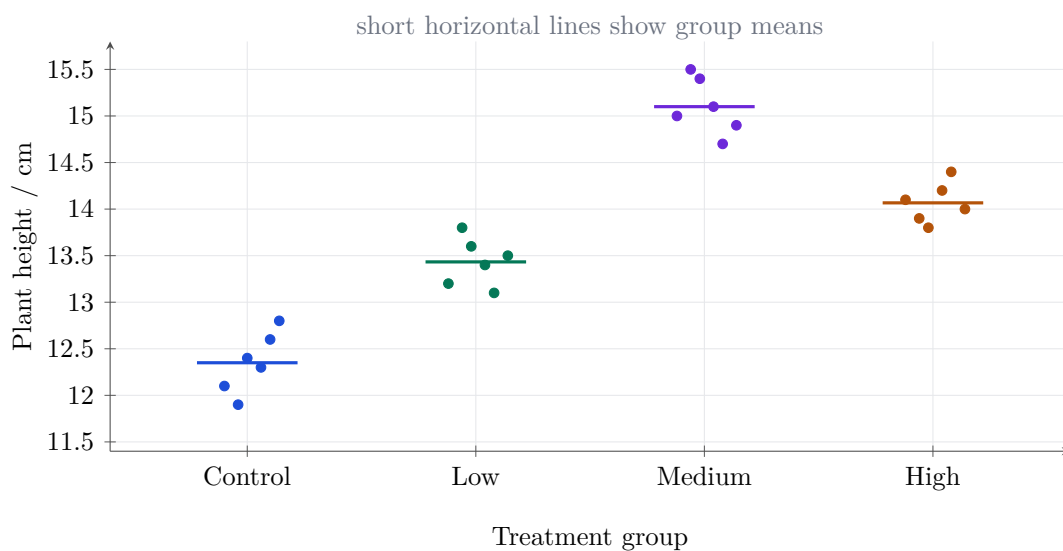
H_A : at least one group mean is different.

What ANOVA can and cannot say

A significant one-way ANOVA says that the group means are not all compatible with one common population mean. It does not, by itself, tell which specific groups differ. For that, use planned contrasts or post-hoc comparisons.

4 Visual first: four groups

Before any p-value, plot the raw data. A raw-data plot often shows whether the group difference is large, whether spread is similar, and whether one outlier is driving the conclusion.



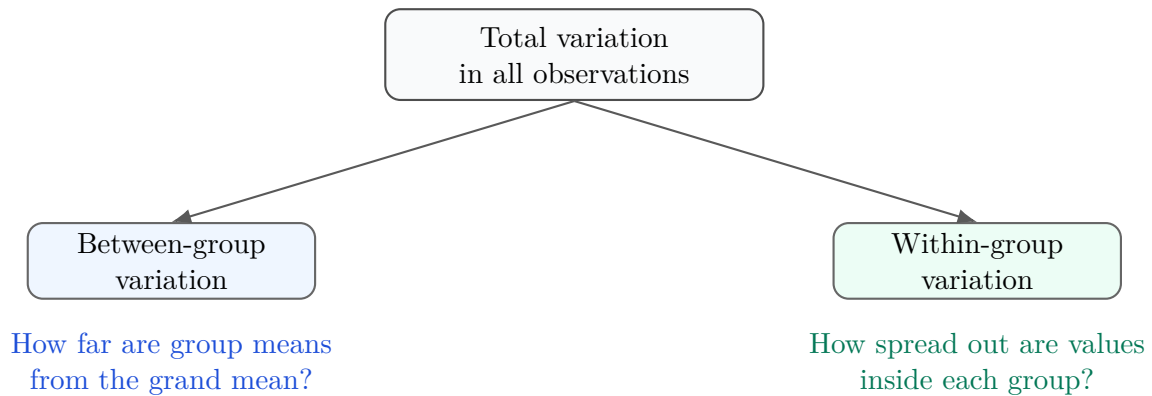
Quick check

From the plot alone, answer cautiously:

1. Which treatment appears to produce the tallest plants?
2. Does the High group look higher than the Medium group?
3. Does the spread look roughly similar across groups?

5 Between-group and within-group variation

ANOVA compares two sources of variation.



The F-statistic is the central test statistic in ANOVA:

$$F = \frac{MS_{\text{between}}}{MS_{\text{within}}}.$$

- If group means are close together compared with within-group variation, F is near 1.
- If group means are far apart compared with within-group variation, F becomes large.

The intuition of F

ANOVA does not ask only whether group means differ. It asks whether the group means differ more than expected relative to the noise inside the groups.

6 Worked example: fertilizer and plant height

A student measures plant height after 14 days under four fertilizer conditions. Each group has six independent plants.

Group	Plant height / cm					
Control	12.1	12.4	12.6	11.9	12.3	12.8
Low	13.2	13.6	13.1	13.8	13.4	13.5
Medium	15.0	15.4	14.7	15.5	15.1	14.9
High	14.1	13.8	14.4	13.9	14.2	14.0

The descriptive statistics are:

Group	n	Mean / cm	SD / cm
Control	6	12.35	0.33
Low	6	13.43	0.26
Medium	6	15.10	0.30
High	6	14.07	0.22

A one-way ANOVA gives:

Source	SS	df	MS	F	p
Between groups	23.895	3	7.965	102.00	2.7×10^{-12}
Within groups	1.562	20	0.078		
Total	25.456	23			

Interpretation

The ANOVA result suggests that mean plant height is not the same across all fertilizer groups. If all four population means were equal and the model assumptions were reasonable, an F-statistic this large would be extremely unlikely.

Do not write:

ANOVA proved that fertilizer caused the difference.

A better statement is:

Plant height differed across fertilizer groups. The difference was large relative to within-group variation, and the ANOVA result was statistically significant. The conclusion still depends on the independence of plants, control of growth conditions, and the validity of the experimental design.

7 Effect size for ANOVA

A small p-value says that the observed pattern would be unusual under the null model. It does not tell how large or important the effect is.

For ANOVA, a common effect size is eta-squared:

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}}.$$

For the example:

$$\eta^2 = \frac{23.895}{25.456} \approx 0.94.$$

This means approximately 94% of the observed variation in plant height is associated with the treatment groups in this dataset.

A slightly less biased estimate is omega-squared:

$$\omega^2 = \frac{SS_{\text{between}} - df_{\text{between}}MS_{\text{within}}}{SS_{\text{total}} + MS_{\text{within}}}.$$

For the example:

$$\omega^2 \approx 0.93.$$

Effect size needs context

Do not treat $\eta^2 = 0.06$ as automatically small or $\eta^2 = 0.14$ as automatically large. In some biological systems, a small percentage of explained variation may still matter. In classroom experiments, huge effect sizes can also appear because the design is simplified and tightly controlled.

8 Post-hoc comparisons

A significant ANOVA tells us that the group means are not all equal. It does not identify the exact pairs responsible for the result.

After a significant ANOVA, students may ask:

Which groups differ from which groups?

That is the job of **post-hoc comparisons**, unless the comparisons were planned in advance.



Common post-hoc options include:

Method	When it is useful
Tukey HSD	Comparing all pairs while controlling family-wise error. Often the default after one-way ANOVA.
Dunnett test	Comparing several treatments only against a control, not every pair. Useful in dose-response studies.
Bonferroni correction	Simple correction for a small number of planned comparisons. Conservative if many comparisons are made.
Planned contrasts	Specific comparisons chosen before seeing the data, such as Control vs average of all treatments.

Practical rule

If the research question is “which dose is better than the control?”, do not automatically compare every group with every other group. Use comparisons aligned with the scientific question.

9 ANOVA assumptions

Classical one-way ANOVA depends on several assumptions.

Assumption	Meaning
Independent observations	One plant, culture, patient, or sample should not determine another. This is a design issue, not a software issue.
Continuous response variable	The measured outcome should be numeric and approximately continuous, such as height, mass, reaction rate, or concentration.
Approximately normal residuals	The errors around group means should not be extremely skewed, especially for small samples.
Similar variances	The groups should have reasonably similar spreads. If not, Welch ANOVA may be better.

The assumptions are about the model, not about making the data look pretty. In practice, inspect raw plots and residuals, and think about the experiment.

Do not over-ritualize normality tests

A Shapiro-Wilk test with $p > 0.05$ does not prove normality. With small samples, it may have little power. With very large samples, it may reject harmless deviations. Use plots, design knowledge, and sensitivity checks.

10 When classical ANOVA may not be ideal

Problem	Possible response
Unequal variances and unequal sample sizes	Consider Welch ANOVA instead of classical ANOVA.
Strongly skewed positive measurements	Consider transformation, such as log transformation, if scientifically meaningful.
Ordinal data or severe non-normality	Consider Kruskal-Wallis, but understand what it tests.
Small samples with uncertain assumptions	Plot raw data; consider permutation methods; avoid strong claims.
Counts or proportions	Do not force ANOVA. Consider binomial, Poisson, negative binomial, chi-square, or logistic-type methods depending on the design.

10.1 Kruskal-Wallis is not simply “non-parametric ANOVA”

The Kruskal-Wallis test compares groups using ranks. It can be useful when classical ANOVA assumptions are doubtful, but it does not automatically solve every problem.

- It is less directly about differences in means.
- It still assumes independent observations.
- If distributions have very different shapes, interpretation becomes harder.
- It still needs post-hoc comparisons if the overall result is significant.

Better than test-name thinking

Do not teach students to memorize “normal = ANOVA, non-normal = Kruskal-Wallis”. Teach them to ask what kind of data they have, what claim they need, and whether the conclusion is robust under plausible analysis choices.

11 How to write ANOVA results

A weak IB-style result statement is:

The ANOVA showed a significant difference, so the hypothesis was correct.

A stronger statement is:

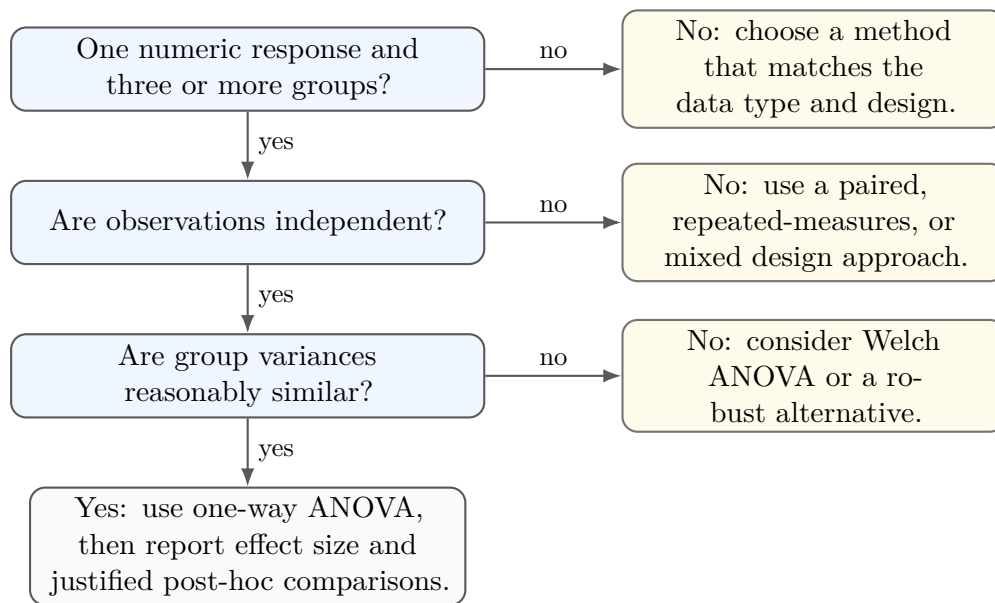
Plant height differed across the four fertilizer groups, $F(3, 20) = 102.00$, $p < 0.001$. The treatment factor explained a large proportion of the observed variation in this dataset, $\eta^2 \approx 0.94$. The Medium fertilizer group had the highest mean height, while the High group was lower, suggesting that additional fertilizer may not continue increasing growth. This conclusion depends on the independence of plants and consistent growth conditions across groups.

Reporting formula

A good ANOVA conclusion should contain:

1. the observed pattern in ordinary language;
2. the test result, such as $F(df_1, df_2)$ and p ;
3. an effect size;
4. post-hoc or planned-comparison results if used;
5. a cautious limitation linked to design or assumptions.

12 Mini decision guide



Decision logic

This guide is not a substitute for judgement. It only forces the correct first questions: data type, independence, spread, and the match between the method and the scientific claim.

13 Quick exercises

Exercise 1: Choose the method

For each case, state whether one-way ANOVA is appropriate, questionable, or inappropriate.

1. Mean reaction rate is measured at 20 °C, 30 °C, 40 °C, and 50 °C, with five independent trials per temperature.
2. Students compare survival counts: alive/dead organisms under four treatments.
3. A student measures the same five plants before treatment, after 7 days, and after 14 days.
4. Absorbance is measured for four different concentrations, and the aim is to build a calibration curve.

Exercise 2: Multiple testing

An experiment has five treatment groups. A student wants to perform every possible pairwise t-test at $\alpha = 0.05$.

1. How many pairwise comparisons are there?
2. Why is this risky?
3. What should the student do instead?

Exercise 3: Interpret a result

A one-way ANOVA gives $F(3, 20) = 6.42$, $p = 0.003$, and $\eta^2 = 0.49$.

1. What does the p-value suggest?
2. What does $\eta^2 = 0.49$ suggest?
3. What can the ANOVA not tell you by itself?

14 Short answer key

1. Exercise 1: (1) appropriate if assumptions are reasonable; (2) inappropriate for ANOVA because the outcome is categorical count data; (3) questionable because repeated measurements on the same plants are not independent, so repeated-measures or paired methods are needed; (4) usually regression, not ANOVA, because the main question is a relationship between concentration and absorbance.
2. Exercise 2: Five groups give $5 \times 4/2 = 10$ pairwise comparisons. The false-positive risk accumulates. Start with an overall model such as ANOVA, then use planned or corrected post-hoc comparisons.
3. Exercise 3: The result would be unlikely under equal population means, assuming the model is appropriate. The group factor explains a large fraction of observed variation in the dataset. ANOVA alone does not identify which specific groups differ.

15 Unit 7 takeaway

Final message

ANOVA is not a button for getting a significant result. It is a way of asking whether group membership explains variation beyond the ordinary scatter inside groups. The correct workflow is: plot the data, compare the full design, report effect size, use post-hoc comparisons only when justified, and write a cautious scientific claim.

Minimal report template

Observed pattern: The response variable differed across the tested groups, with the highest mean in _____ and the lowest mean in _____.

Statistical evidence: A one-way ANOVA indicated that the group means were not all equal, $F(_, _) = _, p = _$.

Effect size: The treatment factor explained approximately _____ of the observed variation, using η^2 or ω^2 .

Cautious conclusion: The result suggests _____, but the strength of the claim is limited by _____.

Final self-check

Before reporting ANOVA, check that you have:

1. plotted the raw data;
2. reported group means and spread;
3. stated the null hypothesis correctly;
4. reported F , degrees of freedom, and p ;
5. reported an effect size;
6. used post-hoc comparisons only when they answer the research question.